

Program 5 : Demonstrate CI Integration by running jenkins in a docker container

project Structure

```
.  
├── docker-compose.yml  
└── Dockerfile
```

Step 1: Create a Dockerfile

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab5$ cat Dockerfile  
FROM jenkins/jenkins:lts  
  
USER root  
  
RUN apt-get update && apt-get install -y docker.io  
  
USER jenkins  
  
.....
```

Step 2 : Create a Docker-compose.yml

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab5$ cat docker-compose.yml  
version: '3.8'  
  
services:  
  jenkins:  
    build: .  
    container_name: jenkins-cli  
    ports:  
      - "8080:8080"  
      - "50000:50000"  
    volumes:  
      - jenkins_home:/var/jenkins_home  
      - /var/run/docker.sock:/var/run/docker.sock  
  
volumes:  
  jenkins_home:
```

3) Build and run the containers

Using Docker File

```
#sudo docker build -t lab5 .
```

```
#sudo docker run -p 8080:8080 -p 50000:50000 --name jenkins_ci lab5 .
```

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab5$ docker compose up
WARN[0000] /home/jayesh-jain/Desktop/DEVOPS_Lab/lab5/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 1.2s (8/8) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 532B                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 140B                                       0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts         0.8s
=> [internal] load .dockerignore                                         0.0s
=> => transferring context: 2B                                           0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:d1ea795c6facd7f549a21 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io     0.0s
=> exporting to image                                                  0.0s
=> => exporting layers                                                  0.0s
=> => writing image sha256:6bb2d917d46148f7e907bf5a48a72a8f1ef8ccef5b642 0.0s
=> => naming to docker.io/library/lab5-jenkins                         0.0s
=> resolving provenance for metadata file                               0.0s
[+] up 3/3
✓ Image lab5-jenkins          Built                                     1.3s
✓ Network lab5_default        Created                                  0.1s
✓ Container jenkins-cli       Created                                  0.1s
Attaching to jenkins-cli
jenkins-cli | Running from: /usr/share/jenkins/jenkins.war
jenkins-cli | webroot: /var/jenkins_home/war
jenkins-cli | 2026-01-22 12:11:26.256+0000 [id=1]      INFO    winstone.Logger#logInternal: Beginning extraction from war file
jenkins-cli | 2026-01-22 12:11:27.478+0000 [id=1]      WARNING  o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
jenkins-cli | 2026-01-22 12:11:27.557+0000 [id=1]      INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.5: built: 2025-12-03T22:18:24.732Z; git: 4905d09d7b6801e792d7b73946cf7b66e4be25d6; ivm: 21.0.9+10-LTS
```

4) Go to the browser and open <http://localhost:8080/>

